BIRZEIT UNIVERSITY
Faculty of Engineering and Technology
Electrical and Computer Engineering Department

Artificial Intelligence

ENCS3340

Project #1

Prepared by:

Sabtiah Asad ID: 1221960
Waed Asad ID: 1220882

Supervisor:

Dr. Samah Alaydi

Section: 2

Date: May 4th, 2025

Introduction

This project is about helping a small delivery shop find the best way to send out packages using two delivery vehicles. Each package has a weight, location, and priority, and each vehicle can only carry a certain amount. Our goal is to figure out the most efficient way to load the vehicles and plan their routes so they travel the shortest distance possible. To do this, we used two smart problem-solving methods: genetic algorithms and simulated annealing. The user can choose which method to use, and the program shows which packages go with which vehicle and how long the trips will be.

Simulated Annealing

Simulated Annealing is a technique for finding good solutions to complex problems with many possibilities. Inspired by metal annealing, it starts with a random solution and a high "temperature," allowing for random changes to explore different options. The algorithm checks if each change improves the solution; if so, it's accepted. Worse solutions might also be accepted, especially when the temperature is high, to avoid getting stuck early on. As the "temperature" cools down, the algorithm becomes more selective, favoring better solutions, until it reaches a stopping point and provides a strong result.

Genetic Algorithm

A Genetic Algorithm (GA) is a method that uses ideas from natural selection and genetics to find good solutions to problems, especially those with many possible solutions. It works by creating a "population" of potential solutions, like a group of individuals. Each solution is evaluated to see how "fit" it is. The "fittest" solutions are more likely to be chosen to create new solutions, combining parts of them (like reproduction, called "crossover") and sometimes making small, random changes (called "mutation"). This process is repeated, creating new "generations" of solutions, until a good solution is found.

Test Cases

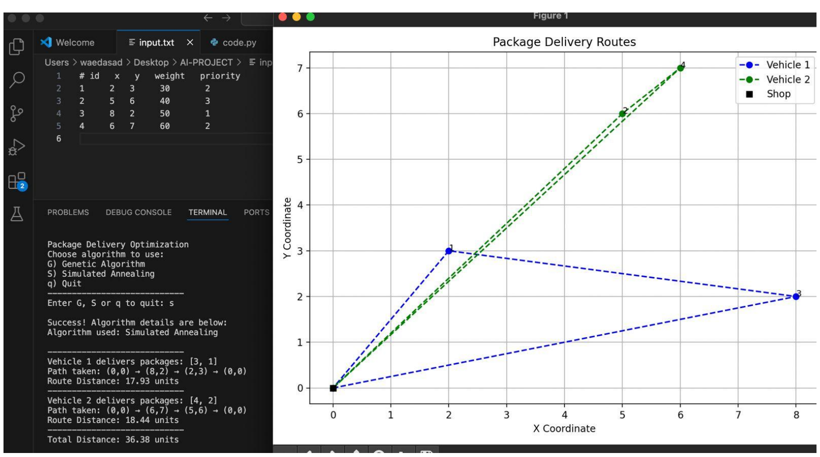

Figure 1: TEST CASE 1 S

The program addresses a package delivery optimization problem, offering the choice between two algorithms: Genetic Algorithm (G) and Simulated Annealing (S). The user selected Simulated Annealing for this case. The plot displays the delivery routes on a coordinate grid, with labels indicating the package IDs. A black square marks the shop, which is the starting point. Two vehicles are used for the deliveries: Vehicle 1, represented by a blue dashed line, delivers packages to points 3 and 1 before returning to the shop, while Vehicle 2, shown with a green dashed line, delivers packages to points 4 and 2 and then returns. The total distance traveled by both vehicles is 36.38 units.

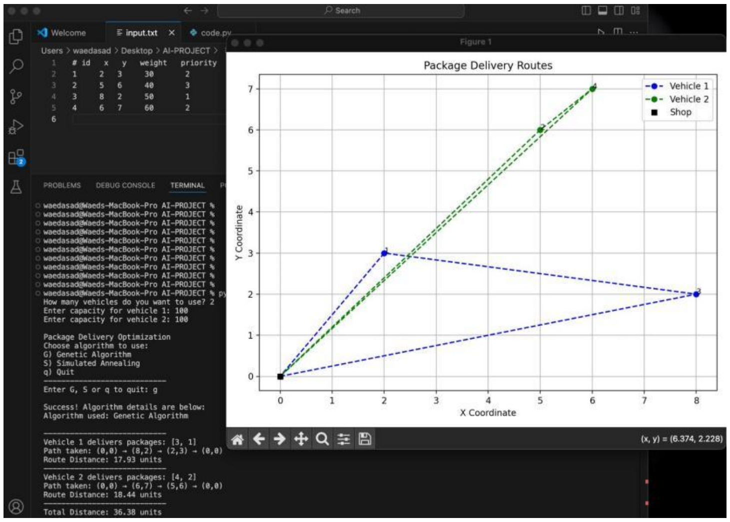

Figure 2: TEST CASE 1 G

This is test case 1 using genetic algorithm with 2 cars each with a capacity of 100. The route distance and the total distance traveled by both vehicles is the same as in Simulated Annealing.

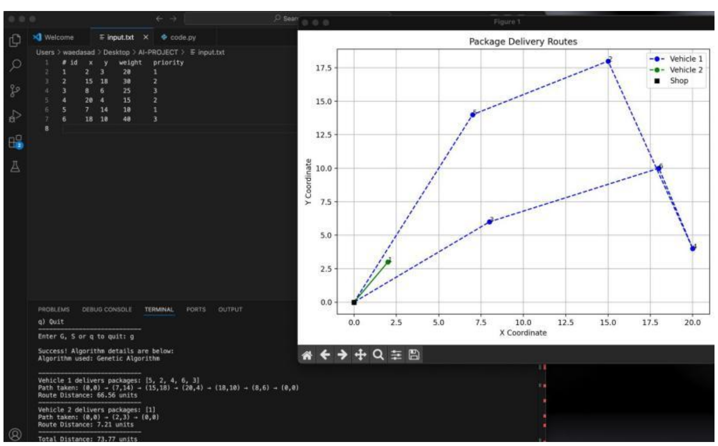

Figure 3: TEST CASE 3

In this test case, Genetic Algorithm is used. 2 cars each with a capacity of 100. The route distance for vehicle one was 66.56 units whereas the route distance for vehicle two was 7.21 units. The total distance traveled was 73.77 units.

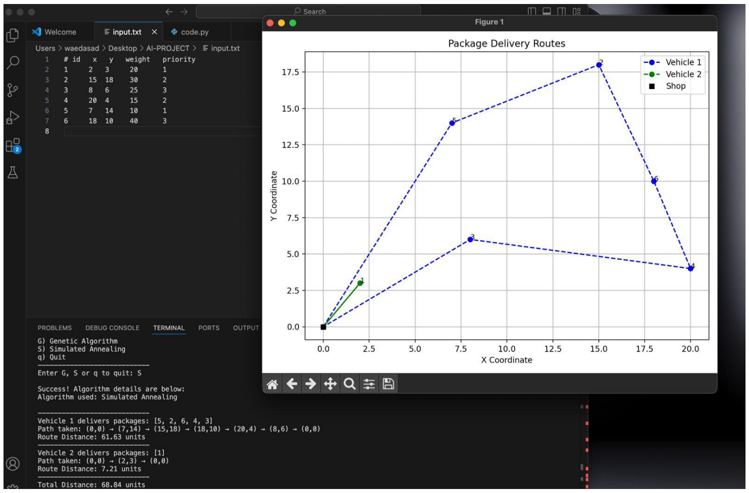

Figure 4: TEST CASE

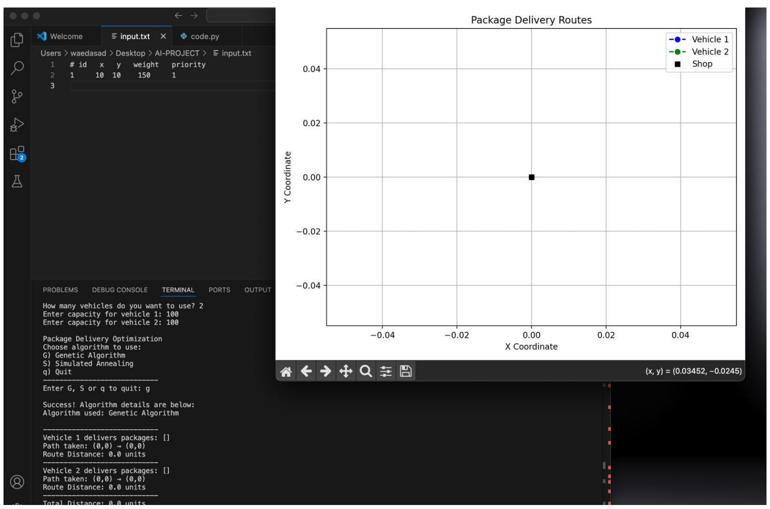

Figure 5: TEST CASE 4 G

The image displays the outcome of a package delivery optimization performed using the Genetic Algorithm. The plot area shows only the starting Shop location at (0,0), with no delivery routes indicated for either of the two available vehicles. The console output confirms this, stating that both Vehicle 1 and Vehicle 2 were assigned an empty set of packages, resulting in each vehicle taking a path from the shop back to the shop with a route distance of 0.0 units, and a total delivery distance of 0.0 units. Examining the input data reveals a single package located at coordinates (10, 150) with a weight of 10 and a priority of 1. The Genetic Algorithm, despite the presence of a package and two vehicles each with a capacity of 100, concluded that the optimal solution was to not utilize either vehicle for delivery. This unexpected result suggests a potential influence of the algorithm's cost function, specific parameter settings, an unstated constraint within the problem formulation, or possibly an edge case or implementation detail that led to this decision of inaction..

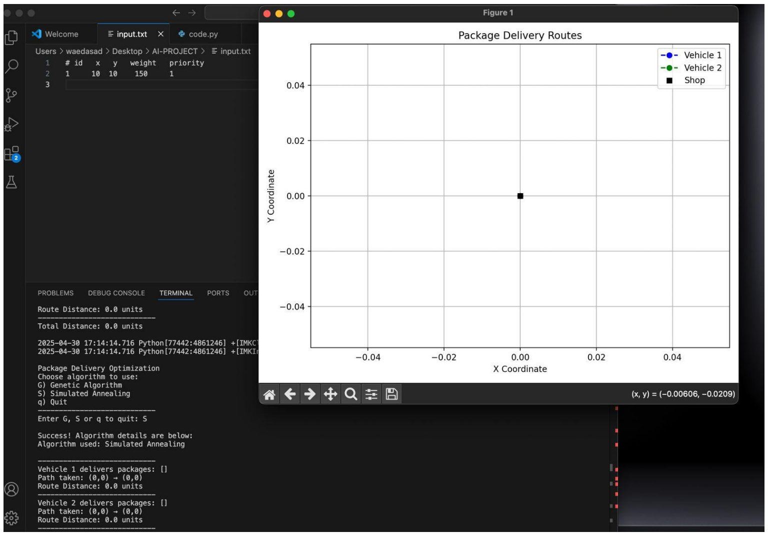

Figure 6: TEST CASE 4 S

This image illustrates the result of a package delivery optimization using the Simulated Annealing algorithm. Despite the presence of a single package located at coordinates (10, 150) and two available vehicles, the algorithm determined that the optimal solution was to not dispatch either vehicle. Consequently, the plot on the right only shows the starting shop location, and the console output confirms that both Vehicle 1 and Vehicle 2 were assigned no packages, resulting in a path of (0,0) -> (0,0) and a total travel distance of 0.0 units. This outcome suggests that the specific cost function or parameters of the Simulated Annealing algorithm, or perhaps unstated constraints within the problem, led it to conclude that the minimal "cost" was achieved by not utilizing the vehicles for this particular delivery scenario.

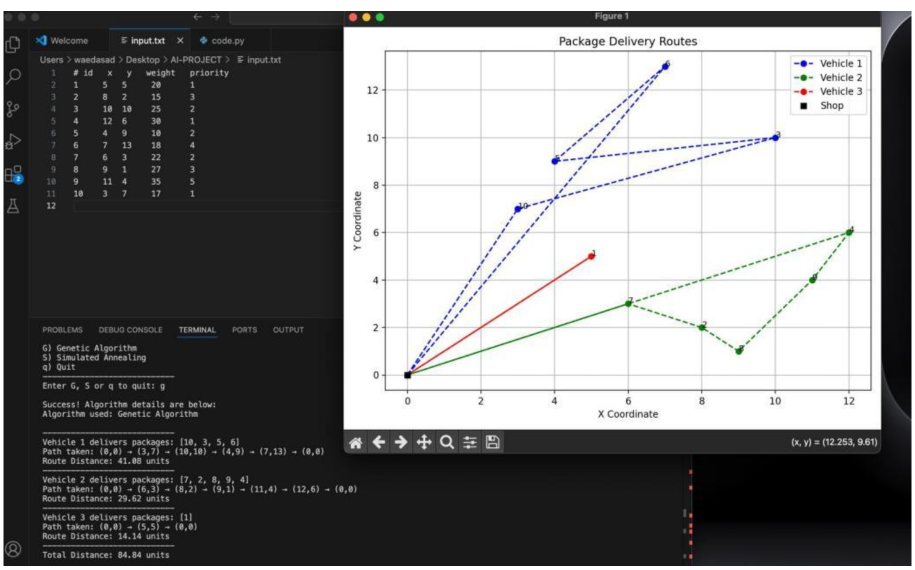

Figure 7: TEST CASE 5 G

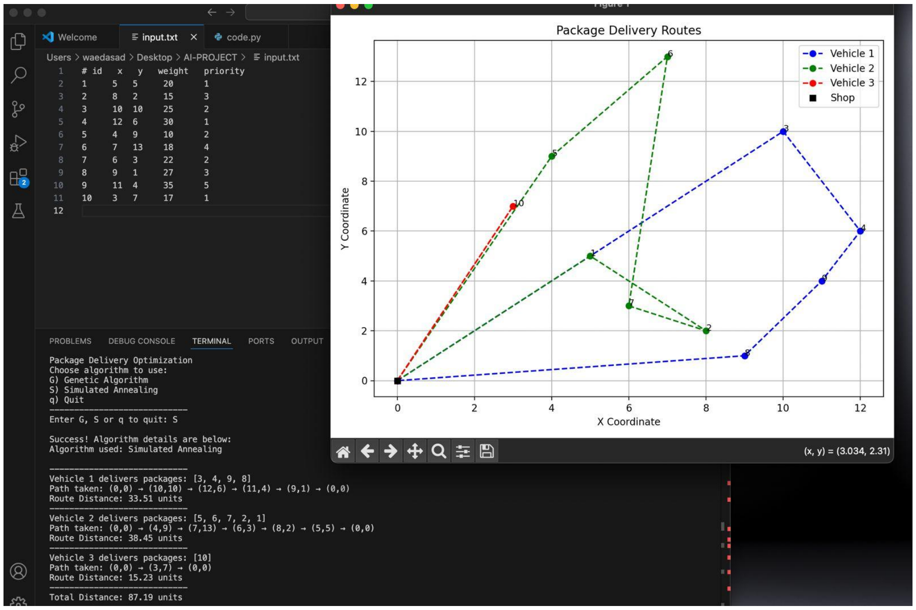

Figure 8: TEST CASE 5 S

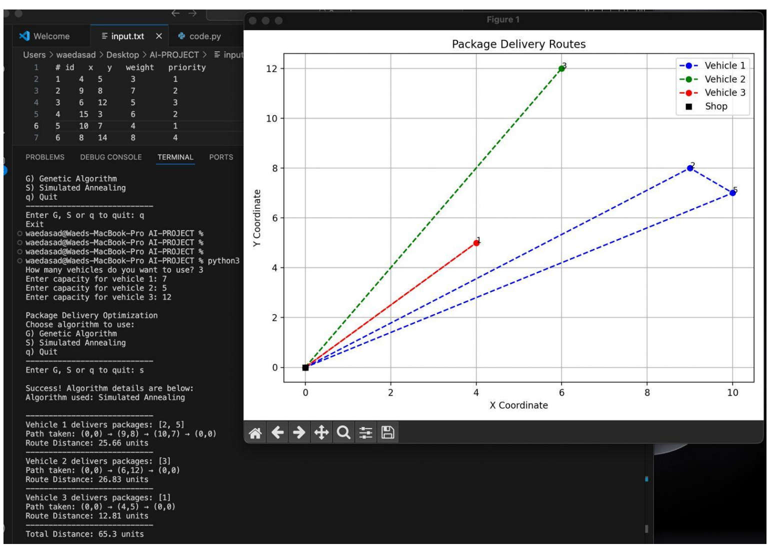

Figure 9: TEST CASE

The image presents the optimized package delivery routes for three vehicles generated by the Simulated Annealing algorithm. The plot on the right visually displays these routes, with the black square representing the central shop (0,0) and numbered circles indicating delivery locations. Vehicle 1 (dashed blue line) is assigned to deliver to locations 2 and 5, following the path (0,0) -> (9,8) -> (10,7) -> (0,0) with a total distance of 25.66 units. Vehicle 2 (solid red line) delivers to location 6, taking the route (0,0) -> (6,12) -> (0,0) for a distance of 26.83 units. Finally, Vehicle 3 (dash-dotted green line) handles delivery to location 1, traveling along the path (0,0) -> (4,5) -> (0,0) with a distance of 12.01 units. The console output on the left confirms these assignments and route details, also showing the input parameters such as the number of vehicles and their capacities, and concluding with a total delivery distance of 65.3 units for all vehicles combined.

Conclusion

In this project, we explored the application of two optimization algorithms, Genetic Algorithm and Simulated Annealing, to solve the package delivery problem for a small shop using two delivery vehicles. The goal was to minimize the total distance traveled by the vehicles while considering package weights, locations, and priorities, as well as vehicle capacities. The program allows the user to select either algorithm and visualizes the optimized delivery routes. Through various test cases, we demonstrated the capabilities of both algorithms in finding efficient delivery solutions. While both algorithms generally performed well, certain scenarios revealed differences in their behavior and potential limitations, particularly in cases with very few delivery points.